



Ana Catarina Ribeiro

Ph.D. Candidate in Computer Science — NOVA School of Science & Technology (NOVA FCT)

Ph.D. candidate at NOVA LINES working on automating microservice testing in a black-box setting through formal API specifications. Research combines model checking (TLA+/TLC) with executable, first-order-logic contracts (GLACIER) to close the semantic gap in API testing and address the test-oracle problem, within the ICEPICK framework. Advised by Prof. Carla Ferreira.

CONTACT

Lisbon, Portugal

acm.ribeiro (at) fct.unl.pt

[Google Scholar](#)

[LinkedIn](#)

[GitHub](#)

[Personal Website](#)

RESEARCH INTERESTS

Software testing & verification (black-box)

Microservices & REST APIs

Specification-based & automated testing

Design by contract

Formal methods, TLA+

TECHNICAL SKILLS

Languages: Java, Python, Kotlin, TypeScript, Scala

Formal methods: TLA+/TLC, GLACIER

LANGUAGES

Portuguese — Native

English — Fluent

EDUCATION

Doctoral Programme in Computer Science

Mar. 2021 – present

NOVA School of Science and Technology (NOVA FCT), Universidade Nova de Lisboa

Automating the microservice testing process by leveraging API specifications, formal contracts (GLACIER) and model checking (TLA+/TLC), targeting causality between concurrent operations and data consistency in distributed settings.

Integrated Master's in Informatics Engineering

Sep. 2015 – Feb. 2021

NOVA School of Science and Technology (NOVA FCT), Universidade Nova de Lisboa

Dissertation: "Invariant-Driven Automated Testing" — automation of microservice testing through a dedicated testing tool and specification language.

PUBLICATIONS

- Ribeiro, A. C., Mamede, M., & Ferreira, C. (2026). Systematic API Testing Through Model Checking and Executable Contracts. 19th IEEE International Conference on Software Testing, Verification and Validation (ICST 2026), Daejeon, South Korea.

PRESENTATIONS

- Correct-by-Construction Distributed Systems and Decentralised Applications workshop (CbC-DS), FLoC 2026 (Jul. 2026).
- Presented "Systematic API Testing Through Model Checking and Executable Contracts" at ICST 2026, Daejeon, South Korea (May 2026).
- TLA+ Community Event, co-located with ETAPS 2026, Torino, Italy (Apr. 2026).

ACADEMIC & TEACHING EXPERIENCE

TEACHING ASSISTANT @ NOVA FCT

- Sep. 2026 – Jan. 2027. Introduction to Programming for Science and Engineering
- Mar. 2026 – Jul. 2026. Design and Analysis of Algorithms
- Sep. 2025 – Jan. 2026. Software Engineering
- Mar. 2025 – Jul. 2025. Programming Languages and Environments
- Sep. 2024 – Jan. 2025. Software Engineering
- Mar. 2024 – Jul. 2024. Object-Oriented Programming
- Sep. 2023 – Jan. 2024. Introduction to Programming
- Mar. 2023 – Jul. 2023. Informatics for Science and Engineering
- Jan. 2023 – Feb. 2023. Transversal Competences for Science and Technology
- Mar. 2022 – Jul. 2022. Informatics for Science and Engineering

SUMMER SCHOOLS & TRAINING

- VTSA Summer School, University of Liège, Belgium (Sep. 2025)
- TAROT Summer School, Università degli Studi di Bergamo, Italy (Jul. 2024)
- DARE Summer School, Vrije Universiteit Brussel, Belgium (Sep. 2023)
- Research Skills and Development course, NOVA Doctoral School (Apr. 2022)

CURRICULUM VITAE

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NOVA FCT / NOVA LINGS

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SUPERVISION & MENTORING

Co-supervisor, with Prof. Carla Ferreira, of MSc thesis topics within the ICEPICK research team at NOVA LINGS.

Sep. 2026 – Jul. 2027

- **Miguel Collaço**
Automated Inference of Invariants and TLA+ Specifications for RESTful APIs
- **Rodrigo Patrício**
Black-Box Testing of MCP Servers: From Interface Quality to Agent Behaviour
- **Afonso Ventura**
Automated Dependency and Contract Inference for RESTful APIs

Finished

- **Letícia Silva**
Model-Checking APIs with Graph Queries (2026)
- **Vladyslav Mykitiv**
RAGBIT: RESTful API Call Sequence Generator with Backtracking (2023)
- **Sara Simões**
Functional Testing of Distributed REST Applications (2021)

ACADEMIC SERVICE & OUTREACH

- Student Representative, INForum 2026, NOVA FCT (Sep. 2026)
- Student Volunteer, TopasLX 26 Programming Competition, NOVA FCT (Apr. 2026).
- Organizer, PhD Workshop, NOVA LINGS, NOVA FCT (Oct. 2024)
- Activity Coordinator, Coding Fest, EXPO FCT, NOVA FCT (Apr. 2024)
- Student Volunteer, ICSE 2024, Lisbon, Portugal (Apr. 2024)
- Education Volunteer, Ciência Viva, NOVA FCT (Jul. 2022)
- Student Volunteer, Coding Fest, EXPO FCT, NOVA FCT (Apr. 2022)